WHY THE MURALS ARE DEGRADING



Caves, natural or man-made, have special microclimatic conditions that support a whole ecosystem. The organic mud plaster is Ajanta's Achilles heel, with rainwater seepage, humidity and its fluctuations breeding a rich microbiotic and insect life, and bats and humans contributing to its decay

Rough plaster
9.3-54.1 mm

Fine plaster
0.1 mm

Lime wash
Nanosize

Paint film
0.1 mm

Bats

Centuries of bat guano still wreak havoc, having penetrated the plaster and rock surfaces, its urea feeding the chain of life



Components in mud plaster

Clay, with a blend of small proportions of ferruginous earth, rock grit, sand and up to 14% organic matter (vegetable fibre, paddy husk) that supports a chain of life, which has devoured 25% of Ajanta's murals. Ellora's plaster included hemp, strikingly helping it avoid such biodeterioration

Components in paint film

Red, yellow and green ochres from local basaltic rock; lampblack or soot; kaolin clay and lime for white; lapis lazuli from Afghanistan for blue. These are mixed with organic glue or binder made of animal fat

Light

Tungsten, mercury and halogen lit Ajanta for years, their heat fading the colours. Fibreoptics, LED are cooler, but pigments are also sensitive to light

Sources: Various studies by the ASI, Science branch, and the Tourist Environmental Impact Assessment Study by CSIR-NEERI, 2013-14, submitted in 2015